

Spectral Analysis Report

Molecule: Benzene (Final Test)

IUPAC Name: Benzene

Common Name: Benzene, Benzol, Phene

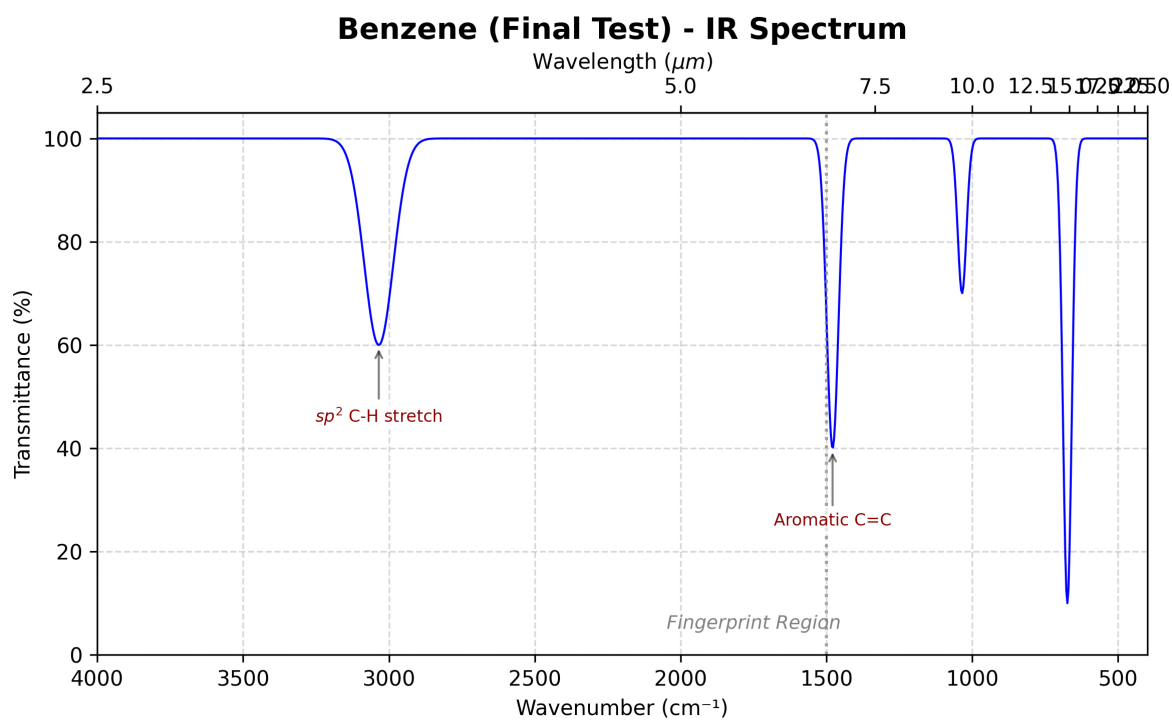
Molecular Formula: C_6H_6

Date: 2026-03-02 06:29:05

Molecular Structure



Infrared Spectrum (IR)

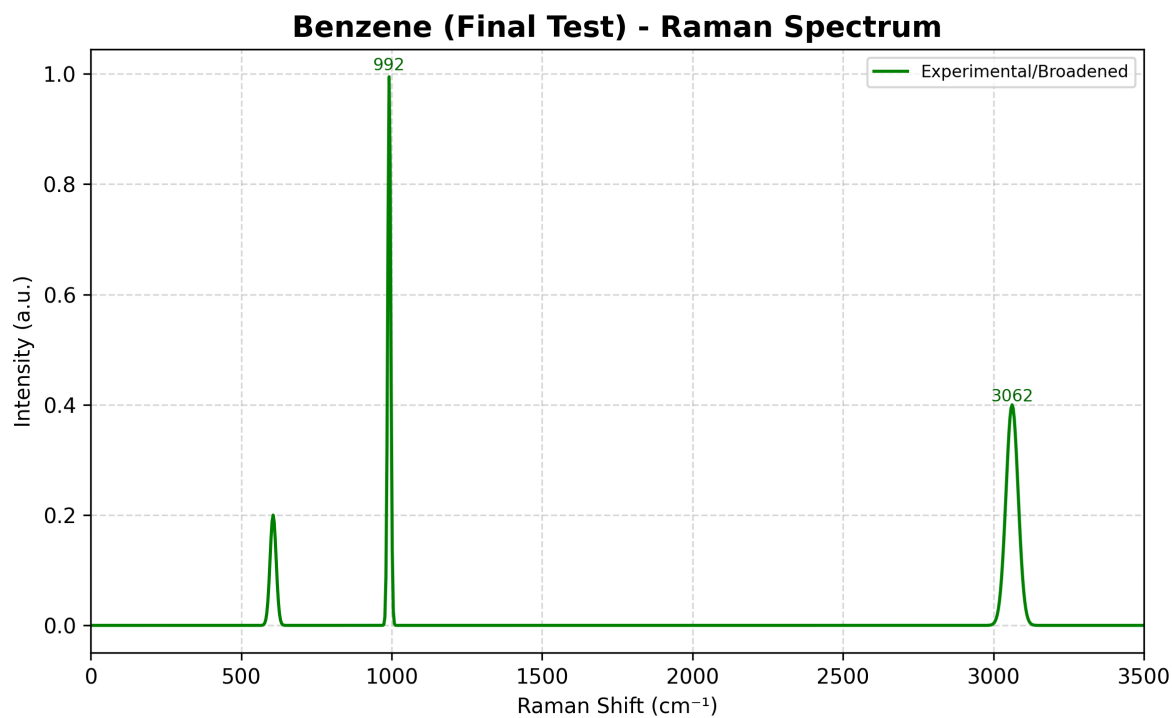


Peak List:

Frequency / Shift	Intensity
3035.00	60.00
1479.00	40.00
674.00	10.00

Interpretation: Corrected: No C=O peak. Major peaks at 3035 (C-H) and 674 (C-H out-of-plane).

Raman Spectrum

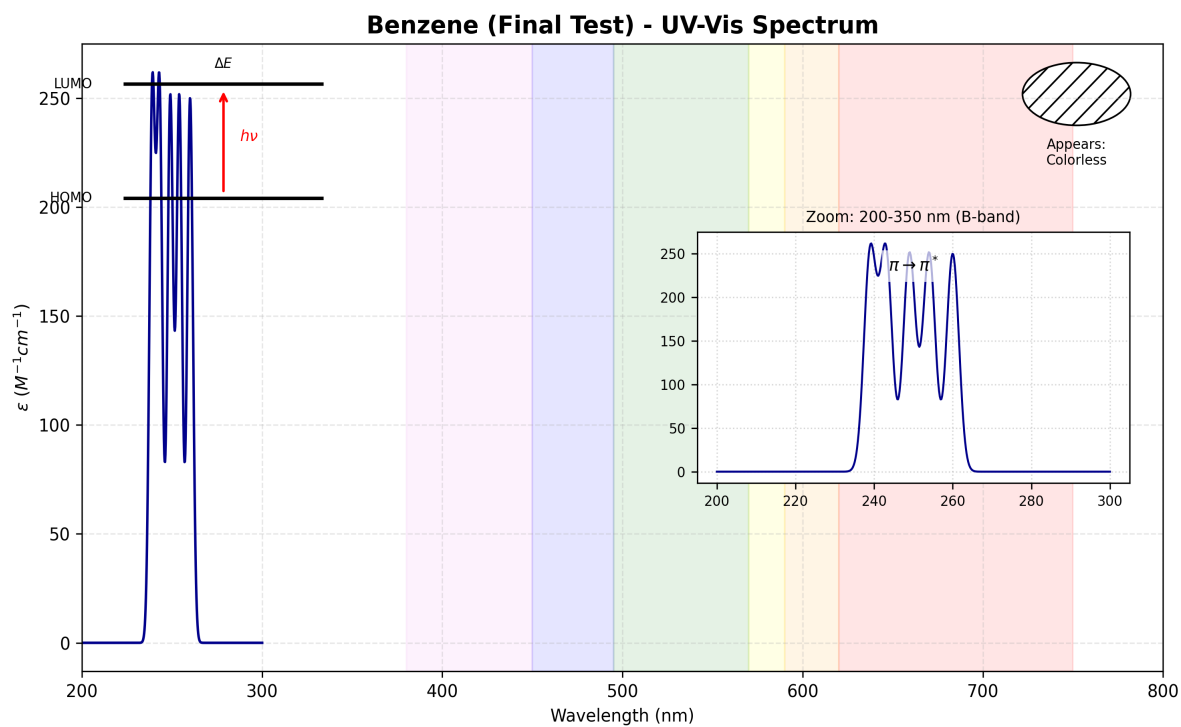


Peak List:

Frequency / Shift	Intensity
3062.00	0.40
992.00	1.00

Interpretation: Dominant ring breathing mode at 992 cm-1 (VS).

UV-Vis Spectrum

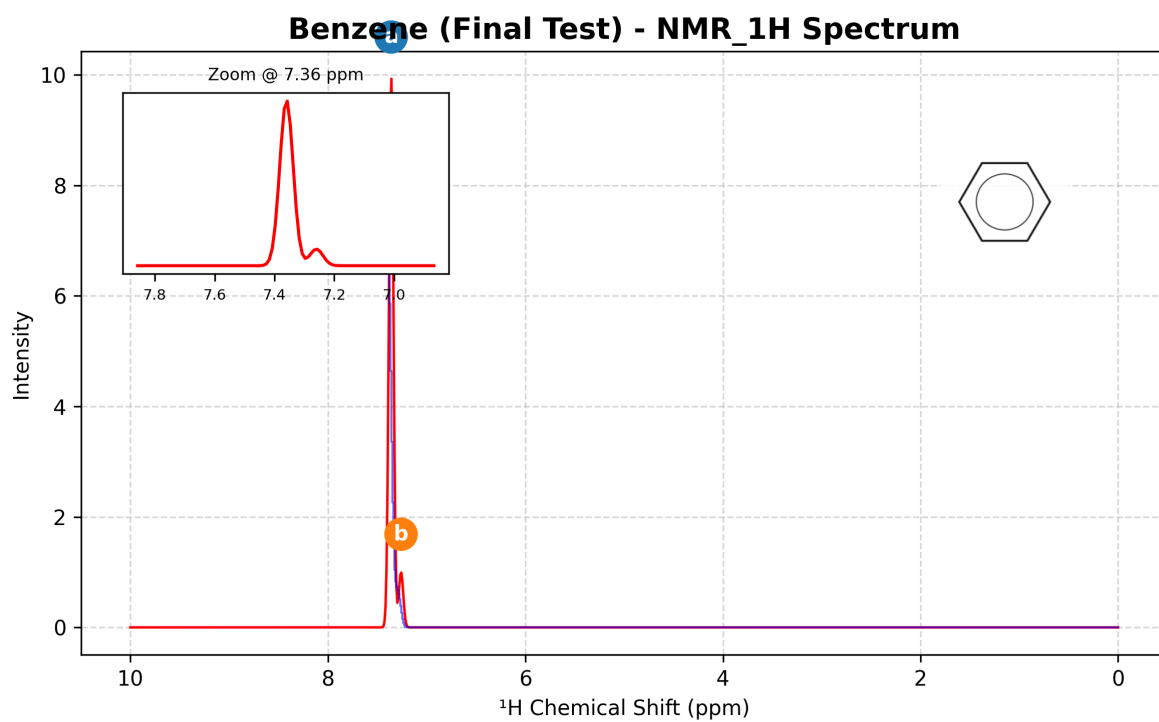


Peak List:

Frequency / Shift	Intensity
254.00	0.20

Interpretation: Characteristic fine structure (B-band) centered at 254 nm.

^1H NMR Spectrum

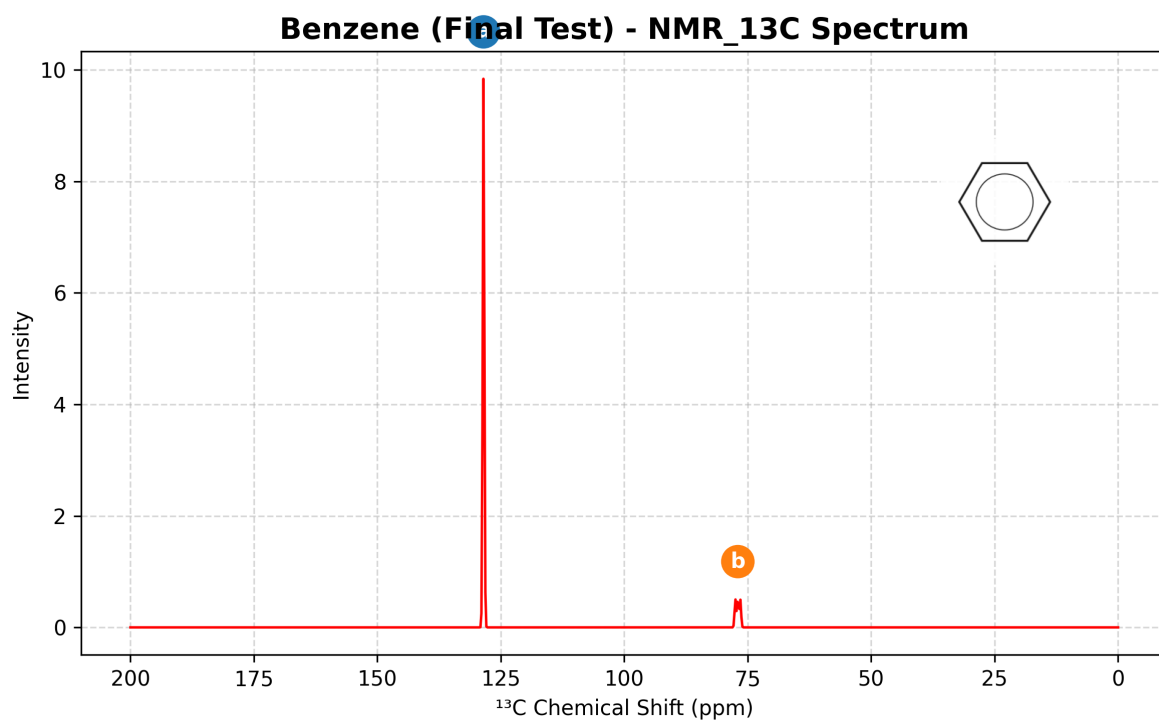


Peak List:

Frequency / Shift	Intensity
7.36	10.00
7.26	1.00

Interpretation: Benzene singlet at 7.36 ppm. Residual CDCl_3 at 7.26 ppm.

^{13}C NMR Spectrum



Peak List:

Frequency / Shift	Intensity
128.50	10.00
77.00	0.50

Interpretation: Benzene carbon at 128.5 ppm. Solvent triplet at 77 ppm.